Operational Research and Optimization/

Group Project

Today's Agenda:

1. Problem Overview
2. Input Data Handling
3. Model Building
4. Team

Case #1

Sophie's Module Selection

Problem Overview

Nature of the Problem

This is an Integer Programming (IP) optimization problem.

The decision is discrete (select or not select a section) therefore, it is a binary integer programming problem.

Sophie must select 5 modules for final year while

- Following university rules
- Avoiding timetable clashes
- Avoiding morning classes
- Maximising her overall satisfaction (rating)

Problem Overview

Beneficiary

#1

Sophie (Student):

Secures the best possible learning experience, avoids schedule conflicts, and fulfills university graduation requirements.

#2

University Administration:

Ensures that students enroll in the correct mandatory courses, aiding in better academic planning and resource allocation.

#3

Professors & Departments:

Helps efficiently manage course enrollments, preventing classes from becoming overcrowded or underfilled.

Problem Overview

Objective

Sophie wants to **select the best combination of five module sections that maximizes overall rating.**

Each section has a rating based on:

- Content (most important)
- Instructor reputation
- Timing (least important)

The goal is to

maxTotal Rating

Input Data Handling

Parameters

	Module(i)	Time Slot (j)	Rating (rij)
Business Strategy	BS101	1	4.3
Business Strategy	BS101	2	3.8
Business Strategy	BS101	3	3.5
Business Strategy	BS101	5	3.5
Business Strategy	BS101	7	4.6
Business Strategy	BS101	9	2.7
International Finance	FIN300	3	3.5
International Finance	FIN300	9	3.3
Intergenerational Computing	CS101	8	4.4
Intergenerational Computing	CS101	10	3.1
Web Design for NonProfit Organizations	CS102	2	3.7
Web Design for NonProfit Organizations	CS102	8	3.5
Data Analysis in Finance	FIN315	4	3
Data Analysis in Finance	FIN315	7	3.7
Risk Management	FIN316	1	3.6
Risk Management	FIN316	6	3.9
Options, Futures and Swaps	FIN317	2	3.2
Options, Futures and Swaps	FIN317	9	3.4
Fixed Instruments and Markets	FIN318	1	3
Fixed Instruments and Markets	FIN318	3	3.5

Time Slot (j)	
1	M 6-8:45 P.M
2	T 6-8:45 P.M
3	W 6-8:45 P.M
4	Th 6-8:45 P.M
5	F 6-8:45 P.M
6	M 1:25-3:15 P.M / W 1:25-2:20 P.M
7	M 1:25-2:20 P.M / W 1:25-3:15 P.M
8	W 2:30-5:15 P.M
9	T 1:25-3:15 P.M / Th 1:25 - 2:20 P.M
10	Th 2:30-5:15 P.M

i : represents the module

j: represents the time slot (only one of 10 time slots)

r_{ij}: represents the rating of module i in time slot j

Input Data Handling

Sets

M : Set of all modules available.

T_i : Set of available sections for module i, where $i \in M$.

r_{ij} : Rating of module i in section j, where $j \in T_i$.

F : Set of finance optional modules = $\{FIN_{315}, FIN_{316}, FIN_{317}, FIN_{318}\}$.

I : Set of industry modules = $\{CS_{101}, CS_{102}\}$.

C : The set containing all pairs of module sections that have conflicting (overlapping) meeting times.

Decision Variables

The model requires binary decision variables to determine if a specific module section is selected:

$$x_{ij} \begin{cases} 1, \text{ if module } i \text{ is chosen in time slot } j \\ 0, \text{ otherwise} \end{cases}$$

Objective Function

Maximise total rating:

$$\max \sum_{i \in M} \sum_{j \in T_i} r_{ij} x_{ij}$$

M : Set of all modules available.

T_i : Set of available sections for module i, where $i \in M$.

r_{ij} : Rating of module i in section j, where $j \in T_i$.

This ensures Sophie selects the highest-rated combination.

Model Constraints

#1 Module Requirements

- Must enroll in BS101:

$$\sum_{j \in T_{BS101}} x_{BS101,j} = 1$$

- Must enroll in FIN300:

$$\sum_{j \in T_{FIN300}} x_{FIN300,j} = 1$$

Model Constraints

#1 Module Requirements

- Must choose exactly one industry-based module:

$$\sum_{i \in I} \sum_{j \in T_i} x_{ij} = 1$$

- Must choose exactly two finance optional modules:

$$\sum_{i \in F} \sum_{j \in T_i} x_{ij} = 2$$

Model Constraints

#2 Scheduling Constraints

- Cannot take more than one section of the same module:

$$\sum_{j \in S_i} x_{ij} \leq 1 \quad \forall i \in M$$

- Cannot take more than one module in the same time slot:

$$\sum_{i \in M} x_{ij} \leq 1 \quad \forall j$$

Model Constraints

#3 Binary Constraint

A module is either taken or not taken:

$$x_{ij} \in \{0, 1\} \quad \forall i \in M, j \in S_i$$

Model Building

Model Constraints

#4

Avoiding Conflicting Time Slots

If two sections are scheduled in conflicting time slots, they cannot both be selected.

$$x_{ij} + x_{kl} \leq 1 \quad \forall\{(i,j),(k,l)\} \in C$$

x_{ij} : The binary decision variable representing whether module i, section j is selected.

x_{kl} : The binary decision variable representing whether module k, section l is selected.

≤ 1 : Ensures that at most one of the two conflicting sections is selected (they cannot both be taken).

$\forall$: "For all" or "for every".

$\{(i,j),(k,l)\}$: A specific pair of module sections.

$\in$: "In" or "an element of".

C : The set containing all pairs of module sections that have conflicting (overlapping) meeting times.

The Team

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Thank you!

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